

Stable Two-dimensional Nanoconfined Ionic Liquids with Highly Efficient Ionic Conductivity

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Amid the burgeoning environmental concerns, electrochemical energy storage is of great demand, inspiring the rapid development of electrolytes. Quasi-liquid solid electrolytes (QLSEs) demonstrate exciting properties that combine high ionic conductivity and safety. Herein, a QLSE system is constructed by confining ionic liquids (ILs) into 2D materials-based membranes, which creates a subtle platform for the investigation of the nanoconfined ion transport process. The highest ionic conductivity increment of 506% can be observed when ILs are under nanoconfinement. Correlation of experimental results and simulation evidently prove the diffusion behaviors of ILs are remarkably accelerated when confined in nanochannels, ascribing from the promoted dissociation of ILs. Concurrently, nanoconfined ILs demonstrate a highly ordered distribution, lower interplay, and higher free volume compared against bulk systems. This work reveals and analyzes the phenomenon of ionic conductivity elevation in nanoconfined ILs, and offers inspiring opportunities to fabricate the highly stable and efficient QLSEs based on layered nanomaterials for energy storage applications.

1. Introduction

The excessive and accelerated depletion of existing fossil fuels has created a global energy and environmental crisis, promoting the fast-growing demand for various energy storage systems. However, a series of material limitations have seriously hampered the development of the current energy

storage devices.^[1–3] Specifically, electrolytes as the major component leading to batteries failures can hardly satisfy the expanding requirements for the renewable energy industry and its products, e.g. electric vehicles.^[4–6] The three widely used electrolytes, i.e., organic, aqueous, and solid-state electrolytes all have their own defects. Being the most common, organic solvents, are mostly volatile and highly flammable.^[7] The undesirable oxidative decomposition^[7,8] and the potential detrimental reactions^[9] seriously restrict the electrochemical performance of high-voltage batteries. Compared with organic, aqueous electrolytes demonstrate a much higher conductivity, but the narrow thermodynamic stability window impedes the extensive applications of aqueous electrolytes.^[10] Solid-state electrolytes are exempted from most safety concerns, but their low room-temperature conductivity restricts the usage.^[11] Consequently, an

innovative electrolyte is urgently needed for developing state-of-the-art energy storage systems.

As an environmentally benign organic substance, ionic liquids (ILs) are an attractive alternative to conventional electrolytes for their noteworthy characteristics, such as relatively high intrinsic conductivity, wide electrochemical operating range, high thermal and chemical stability, low vapor pressure, good capability of dissolving, and nonflammability.^[12,13] However, the far-reaching applications of ILs have been hampered by their high viscosity, lower conductivity than expected, and liquid state.^[14,15] Ideal ILs should entirely consist of nonassociated ions, but in reality, they form ion aggregates and contribute less to conductivity.^[16,17] Meanwhile, the leakage of ILs is another urgent issue. Therefore, the concept of quasi-liquid solid electrolyte (QLSE) has been brought to address this problem.^[18,19] QLSEs cover a combination of the safety provided by the nonflammable solid-state electrolyte together with the inherent advantages of liquid systems,^[20,21] and thus have been reported some remarkable researches in the battery field.^[22] In particular, carbon nanotubes,^[23,24] nanoparticles,^[25,26] nanofibers,^[27,28] and nanoporous materials^[21,29] have been selected as the supporting functional hosts to fabricate QLSEs. In comparison with the nanomaterials above, 2D nanochannels generated by nanosheets are promising candidates for QLSE supporters because of the extended interlayer space created by the layer-to-layer repulsive interactions or intercalated compounds,

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which can accommodate more ions to enhance the flux.^[30–34] Ease of fabrication and functionalization further promotes the applications of 2D nanochannels.^[35–40] Moreover, the effect of nanomaterials on ILs is controversial and likely depends on different confinement scales.^[41] Nanomaterials could depress the viscosity of ILs in bulk systems but show a negative impact in around hundreds-nanometer confinement.^[42,43] The physicochemical and dynamical characteristics of ILs moving through channels comparable its own size (nano to sub-nanometer) remains to be explored.

This work focuses on expounding the mechanism of nanoconfinement-induced ionic conductivity exaltation with some experimental facts. Various nanosheets and ILs are selected here to investigate the confinement effect on ionic conductivity. 2D material nanochannels were constructed by vacuum filtration with ILs imbibition. Necessary characterizations were performed to get an initial understanding of the electrochemical systems. Experimental evidence of enhancement of ionic conductivity under nanoconfinement is firstly presented by electrochemical impedance spectroscopy (EIS). The highest ionic conductivity increment of ILs involved in this work is more than a factor of 5, significantly improving the performance of ILs as electrolytes. Then, convincing explanations were put forwards according to the experimental results and molecular dynamics (MD) simulation. Pulsed-field gradient nuclear magnetic resonance (PFG-NMR) was employed to obtain the diffusion coefficients of ILs in bulk and confined systems, with which results the dissociation level can be calculated. MD simulation analyzed the differences of interplay and free volume between bulk and nanoconfined ILs. These offer a more profound understanding of the conductive properties of ILs in 2D nanoconfinement.

2. Results and Discussion

2.1. Membrane Preparation and Characterization

In order to investigate the subnanometric confinement effect of 2D nanochannels on transport behaviors of ILs, graphene oxide (GO), and molybdenum disulfide (MoS_2) nanosheets with characteristic interlayer spaces were selected here to build the nanometric space. GO nanosheets were prepared according to a common modified Hummers method.^[44] The average lateral dimensions of the obtained GO sheets are in the microscales and the thickness is around 1 nm, respectively (Figure 1a), which is in agreement with the typical thickness of monolayer GO nanosheets. A brown translucent and stable colloidal solution containing GO nanosheets presents an apparent Tyndall effect (Figure 1b), indicating the great dispersion of GO nanosheets in water. XPS spectra confirm the presence of abundant oxygen-containing functional groups on the surfaces of GO nanosheets and the contents of C 1s and O 1s declare the normal oxidation degree of prepared GO (Figure S1 and Table S1, Supporting Information). These oxygen functional groups make the GO dispersion shows a negative charge, as suggested by zeta potential of -40.2 mV at pH = 7 (Table S2, Supporting Information). Simple imidazolium cation, 1-ethyl-3-methylimidazolium $[\text{EMIM}]^+$, accompanied with three widely used anions featuring different sizes and molecular structures, spherical

(tetrafluoroborate, $[\text{BF}_4]^-$), elongated with a delocalized charge (bis(trifluoromethyl sulfonyl)imide, $[\text{NTf}_2]^-$), and elongated surfactant-like (ethyl sulfate, $[\text{EtSO}_4]^-$) were considered and used to explore the transport disciplines of 2D-confined ILs and their chemical structures are shown in Figure 1c. GO membranes (GOMs) were fabricated by vacuum-filtrating a certain volume of GO dispersion on polycarbonate (PC) substrates.

Scanning electronic microscopic (SEM) was exploited to characterize the morphology of GOMs. Apparent wrinkles can be observed on the surface of GOMs owing to the flexible nature of GO nanosheets (Figure 1d). Typical laminated structure of re-stacked 2D materials without large pores or large chunk of disordered flakes can be seen clearly in the cross-section of GOMs, confirming the as-prepared GOMs are well-stacked (Figure 1e). To investigate the wettability of ILs on GOMs, contact angle tests were performed (Figure 1f). All contact angles of ILs on GOMs are below 60° , validating the good wettability of ILs with GO membrane. Based on the Young–Laplace equation, the positive capillary forces guarantee the easy and fast imbibition of IL into GOMs by a simply dropping spin-coating method.^[45] Compared with GOMs, the images of GO-confined-IL membranes (GOILMs) demonstrate rougher surfaces, resulting from the shortened and densified wrinkles (Figure 1g and Figure S2a,b, Supporting Information). The pristine lamellar structure of GOMs is substituted by a smooth and unitary plane because of the full imbibition of ILs and the thickness of GOILMs is remarkably enhanced (Figure 1h, and Figure S2c,d, Supporting Information). The even incorporation of ILs into GOMs is further affirmed by the energy dispersive spectrometer (EDS) mapping images of GOILMs, from which elements of ILs are found uniformly distributed (Figure 1e and Figures S3 and S4, Supporting Information). Similar enhanced phenomenon in thickness has also been found in the X-ray diffraction (XRD) analysis. In contrast to an interlayer distance of 0.83 nm of GOMs, the lamellar spaces of $\text{GO}[\text{EMIM}][\text{BF}_4]$, $\text{GO}[\text{EMIM}][\text{NTf}_2]$, and $\text{GO}[\text{EMIM}][\text{EtSO}_4]$ membranes are expanded and calculated to be 1.73 , 0.99 , 2.46 nm, respectively (Figure 2a), indicating the existence of ILs in the interlayer of GO, and the distinction among the thickness increment of GOILMs is originated from the synergistic effect between the dimensions of anions and the interaction with GO nanosheets.

Spatial confinement can influence the physicochemical properties of IL confined in the GO nanochannels. Herein, attenuated total reflection of Fourier transform infrared spectroscopy (ATR-FTIR) was employed to get a full comprehension of the interplay between IL and GO nanosheets (Figure 2b–d). The characteristic absorption at 1725 and 1621 cm^{-1} of GOMs are corresponding to the functional groups, $\text{C}=\text{O}$ and $\text{C}=\text{C}$, respectively, which can also be found in the ATR-FTIR spectra of GOILMs. At the same time, main IR peaks of ILs are also available in the spectra of GOILMs, especially peaks in the fingerprint region associated with anions, confirming that ILs are immobilized in the GOMs. The two FTIR peaks between 3000 and 3200 cm^{-1} are assigned to the $\text{C}-\text{H}$ stretching vibrational modes, which are corresponding to the cation, $[\text{EMIM}]^+$, of ILs, are mostly red-shifted to some extent for $[\text{EMIM}][\text{BF}_4]$ and $[\text{EMIM}][\text{EtSO}_4]$ due to the $\pi-\pi$ stacking interactions between the imidazolium rings of ILs between GO.^[46] As for $[\text{EMIM}][\text{NTf}_2]$, these cation-characteristic peaks above are largely blue-shifted

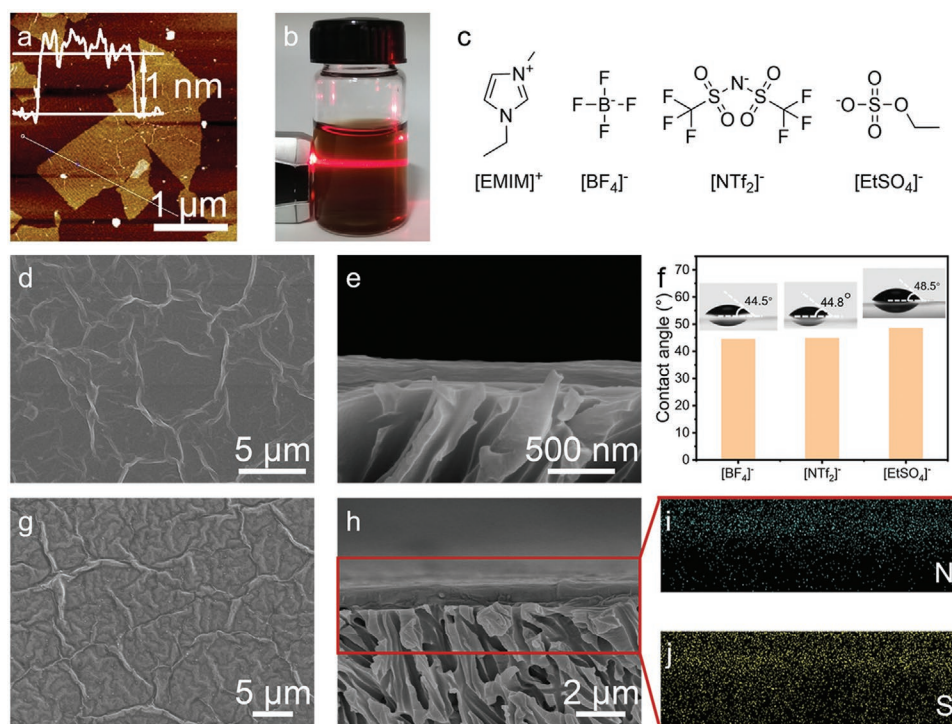


Figure 1. a) Atomic force microscopy (AFM) image of graphene oxide (GO) nanosheets. b) Digital photo of Tyndall effect of GO dispersion. c) Chemical structures of cation and anions of ionic liquids (ILs) used in this work. d) Scanning electronic microscopic (SEM) images of surfaces of GO membranes (GOMs). e) SEM images of cross-sections of GOMs. f) Contact angles of [EMIM][BF₄], [EMIM][NTf₂], and [EMIM][EtSO₄] on GOMs. g) SEM images of surfaces of GO-[EMIM][EtSO₄] membranes. h) SEM images of cross-sections of GO-[EMIM][EtSO₄] membranes. i, j) Energy dispersive spectrometer (EDS) mapping images of N and S elements recorded from the marked zone of Figure 1h, respectively.

since the [NTf₂]⁻ anion display the strongest potential of newly forming hydrogen bond with the hydroxyl groups of GO.^[47] In

comparison to the pure IL, the peaks representing anions of ILs (i.e., 1061 cm⁻¹ of [BF₄]⁻, 1166 cm⁻¹ of [NTf₂]⁻, and 1171 cm⁻¹ of

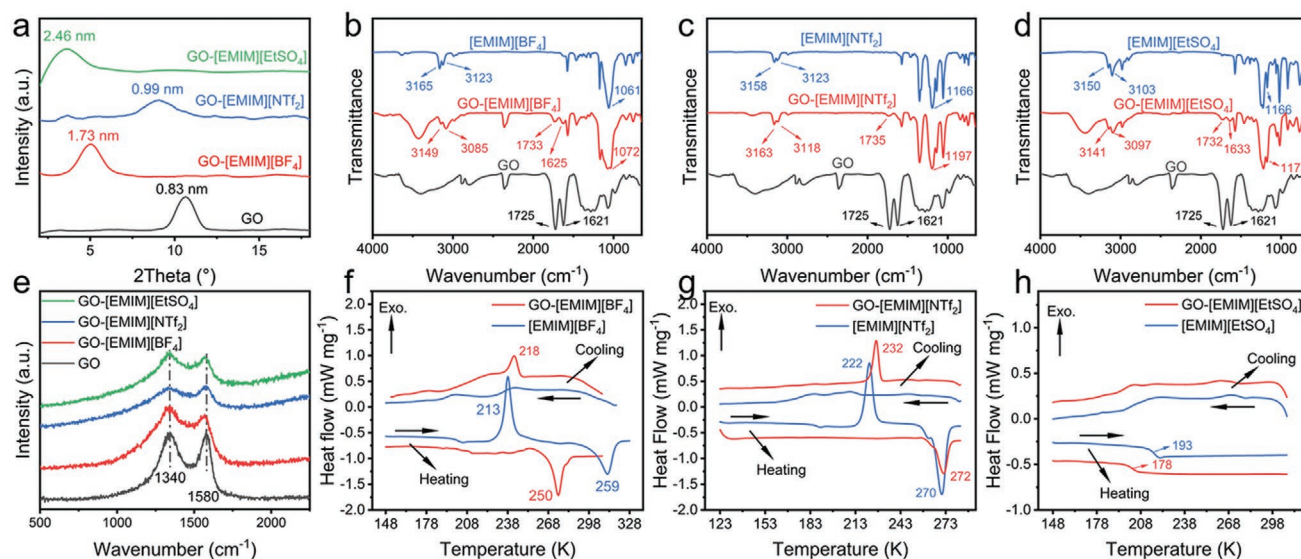


Figure 2. a) X-ray diffraction (XRD) patterns of graphene oxide (GO) GO-[EMIM][BF₄], GO-[EMIM][NTf₂], and GO-[EMIM][EtSO₄] membranes. b–d) Fourier transform infrared spectroscopy (FTIR) spectra of ionic liquids (ILs), GOIL membranes, and GO membrane. b) [EMIM][BF₄], GO-[EMIM][BF₄] membrane, and GO membrane, c) [EMIM][NTf₂], GO-[EMIM][NTf₂] membrane, and GO membrane, and d) [EMIM][EtSO₄], GO-[EMIM][EtSO₄] membrane, and GO membrane. e) Raman spectra of GO, GO-[EMIM][BF₄], GO-[EMIM][NTf₂], and GO-[EMIM][EtSO₄] membranes. f–h) Differential scanning calorimetry (DSC) patterns of ILs and GO-confined-IL membranes (GOILMs), respectively. f) GO-[EMIM][BF₄] membrane and [EMIM][BF₄], g) GO-[EMIM][NTf₂] membrane, [EMIM][NTf₂], and h) GO-[EMIM][EtSO₄] membrane and [EMIM][EtSO₄].

[EtSO₄][−]) demonstrate a consistent shift to higher energy. The observed FTIR shifts verify that both cations and anions can interact with GO nanosheets. The effects of ILs on the structure of GO nanosheets were determined through Raman spectra as shown in Figure 2e. The two representative absorption peaks centered at 1340 (D band) and 1580 cm^{−1} (G band), appeared in neat GOMs and GOILMs with a slight shift, are assigned to the vibration of sp³ carbon atoms from the defects or functional groups in GO structure and the in-plane vibration of sp² bonded carbon atoms.^[48] The relative intensity of D and G bands (I_D/I_G) is a critical parameter to identify the disorder degree of graphene. GOILMs exhibit an increased value of I_D/I_G ratio compared with the pristine GOMs, indicating a higher asymmetric structure of the sp³ defects in GO nanosheets. This may imply the fact that the interplay between the functional groups at the edges of GO nanosheets and ILs is dominant in the GOIL systems.^[49] Thermodynamic properties of nanoconfined ILs were investigated by differential scanning calorimetry (DSC, Figure S8, Supporting Information).

Owing to the unique structural features of ILs, a combination of low-symmetric organic cations and weakly coordinating inorganic or organic anions, the crystallization of [EMIM][BF₄] and [EMIM][NTf₂] is hampered with a glass transition during the cooling period, while an exothermic peak for crystallization occurs when reheating the IL sample (Figure 2f,g). The exceptional phenomenon is known as “cold crystallization,” appearing in the reheating period of some ILs, which mostly depends on the species of ILs.^[50] As for GO-[EMIM][BF₄] and GO-[EMIM][NTf₂] membranes, the crystallization of ILs appears normally in the cooling periods, meaning that when ILs are confined by GO, there is enough space for ionic motion in time to achieve phase transition. This anomalous phenomenon suggests that the free volume of ILs may expand in nanoconfinement space and the arrangement of ILs under nanoconfinement may be more ordered than their counterparts. The melting point of [EMIM][BF₄] moves to a lower temperature when confined by GO, same as major researches,^[15,51] revealing the ability of nanoconfinement to broaden the liquid range of ILs. The slightly increased melting point of [EMIM][NTf₂] may be related to complex hydrogen bonding between anions and GO.^[46,52] For [EMIM][EtSO₄], there is no crystallization throughout the measured temperature range in both bulk and confined conditions (Figure 2h). The glass transition temperature of bulk [EMIM][EtSO₄] is much higher than that under GO-confinement, indicating that nanoconfinement may activate the ionic motion of [EMIM][EtSO₄].^[50] For another 2D material, MoS₂ nanosheets were exfoliated by N-butyl lithium in hexane as described in our previous work.^[53] A stable transparent solution containing nanometer-scale and negatively charged MoS₂ sheets was obtained after centrifuging several times (Figure S5 and Table S3, Supporting Information). Characteristic Raman peaks of 382 and 405 cm^{−1} identify the successful synthesis of the monolayer to bilayer MoS₂ nanosheets (Figure S6, Supporting Information). Ultraviolet visible (UV) absorption spectra of MoS₂ dispersion under thermal treatment further verify that there is no phase transition during the ionic conductivity measurement (Figure S7, Supporting Information). MoS₂-confined-IL membranes (MoS₂ILMs) were prepared by the same methods as GOILMs. Uniform distributions of characteristic

elements of IL are also observed in MoS₂ILMs with an enlarged thickness (Figures S8 and S9, Supporting Information).

2.2. Transport of Nanoconfined Ionic Liquids

The ionic conductivity was measured by an electrochemical workstation with the assistance of an incubator and the nanofluidic device that supports the ion transport through 2D-material-based membranes is displayed in Figure 3a,b and Figure S11 (Supporting Information). EIS was exploited to explore the electrical properties of IL, GOIL, and MoS₂IL systems. A set of representative Nyquist plots of ILs, GOILMs, and MoS₂ILMs at various temperatures are displayed in Figure 3c,d, Figures S12 and S13 (Supporting Information). The ionic conductivity of bulk and confined IL is calculated and plotted as Figure 3e–h shown. All these plots of ionic conductivity demonstrate the same dependence of temperature for both confined and bulk ILs, and the influence of temperature becomes more pronounced under nanoconfinement.

Take GO-[EMIM][EtSO₄] system as an example, when the temperature rises from 288 to 338 K, the ionic conductivity of GO-confined IL is increased by 5.07 S m^{−1} while bulk IL is improved by 0.99 S m^{−1}. The influences of nanoconfinement on ionic conductivity of ILs were investigated through comparing the ionic conductivity of bulk ILs with their confined counterparts. In contrast with three bulk ILs, the ionic conductivities of GO-[EMIM][BF₄], GO-[EMIM][NTf₂], and GO-[EMIM][EtSO₄] membranes are much higher under the whole studied temperatures (Figure 3e–g). The same phenomenon of GO-confinement-induced enhancement of ionic conductivity indicates that the continuous transport channels of GOMs are favorable for the fast transport of IL. Similar nanoconfinement effects have also been found in another nanofluidic system, where MoS₂ nanosheets serve as the nanoconfinement components (Figure 3h). The ionic conductivity of MoS₂-[EMIM][EtSO₄] membranes is more than twice of bulk [EMIM][EtSO₄]. This result declares the universal positive influence of different 2D materials on the ionic conductivity of ILs and the kinds of materials is one of the factors of ionic conductivity enhancement. The conductivity between of different QLSEs reported so far is summarized and listed in Table S4 (Supporting Information), and our 2D material-confined-IL systems exhibit competitively high conductivity among them. GO-confined [EMIM][EtSO₄] system was selected to investigate the practical performance of homemade QLSE (Figure S15, Supporting Information). GO-[EMIM][EtSO₄] membranes, which require much less volume of IL, can exhibit comparable capacitance with bulk [EMIM][EtSO₄] electrolyte at ambient temperature, proving the promising potential applications of QLSE fabricated in this work. XRD and EIS were performed again after 3 months to analyze the stability of 2D material-confined-IL systems. The interlayer spacings of samples demonstrate negligible changes compared with the original membranes (Figures S16 and S17, Supporting Information), indicating the reliability of the 2D material-confined-IL systems. Although the ionic conductivity is slightly decreased after 3 months, it still much higher than that of bulk counterparts, verifying the potential of high-performance energy storage application.

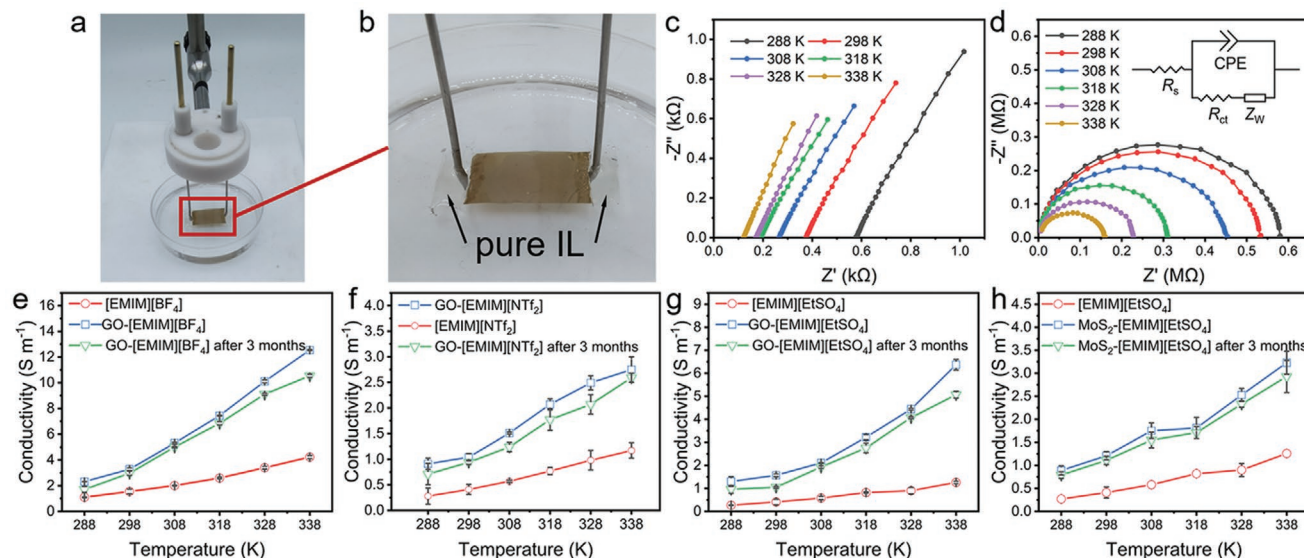


Figure 3. a,b) Nanofluidic device for electrochemical measurement used in this work. c,d) Nyquist plots of graphene oxide (GO) [EMIM][EtSO₄] membrane and [EMIM][EtSO₄] at different temperatures, respectively. The electrochemical measurements were performed at the frequency range from 1 MHz and 100 Hz. e) Ionic conductivity of bulk [EMIM][BF₄], GO-confined [EMIM][BF₄], and GO-confined [EMIM][BF₄] after 3 months. f) Ionic conductivity of bulk [EMIM][NTf₂], GO-confined [EMIM][NTf₂], and GO-confined [EMIM][NTf₂] after 3 months. g) Ionic conductivity of bulk [EMIM][EtSO₄], GO-confined [EMIM][EtSO₄], and GO-confined [EMIM][EtSO₄] after 3 months. h) Ionic conductivity of bulk [EMIM][EtSO₄], MoS₂-confined [EMIM][EtSO₄], and MoS₂-confined [EMIM][EtSO₄].

To further evaluate the promoting effect of nanoconfinement on ionic conductivity, the increment is defined as the ratio of ionic conductivity of bulk and confined IL. And the increments of ionic conductivity at each temperature are listed in Figure S14 (Supporting Information), varying from 163% to 506% within the measured temperature range. The ionic conductivity of bulk and confined IL exhibits a well-fitting Arrhenius-type dependence, from which the activation energy can be calculated (Figure S18, Supporting Information). The activation energy for all samples only varies in a narrow range, resulting from the constant composition of the ionic electrolytes. The slight difference between bulk and confined ILs in various conditions is unable to explain the remarkably enhanced ionic conductivity of confined IL. PFG-NMR was employed to measure the self-diffusion coefficients of bulk and confined IL for the purpose of disassociation assessment.^[54] According to the chemical shift of peaks, diffusion coefficients of cation and anion of IL, GOIL, and MoS₂IL membranes can be obtained from ¹H NMR diffusion ordered spectroscopy (DOSY) spectra performed at 298 K (Figure 4a–c, Figures S19 and S20, and Tables S5–S7, Supporting Information). The diffusion coefficients of [EMIM]⁺ and [EtSO₄][−] in bulk are 2.00×10^{-11} and 1.26×10^{-11} m² s^{−1}, respectively (Figure 4a). Despite the bulkier structure, the faster dynamics of cation is attributed to the homogenous electrostatic field, the flexible alkyl chain, and the planar structure of the imidazolium ring.^[15,55] All of the specific features are conducive to transport. For GO-confined IL, the diffusion coefficients are 5.39×10^{-11} and 4.03×10^{-11} m² s^{−1} for [EMIM]⁺ and [EtSO₄][−], much larger than those of bulk IL (Figure 4b). Additionally, the diffusion coefficient of anion is improved more distinctly than that of cation, which is common under aqueous nanoconfinement systems.^[56] The asymmetry between cation and anion

may be relevant to the chemical characteristics of GO nanochannels. Meanwhile, the diffusion of ions of [EMIM][EtSO₄] is also accelerated under MoS₂ nanoconfinement system (Figure 4c). These improved diffusion behaviors of cation and anion confirm the promoting effect of nanoconfinement on ion transport of nanoconfined IL. And the enhancement is determined by the materials used to build the ion channels. The molar conductivities of bulk, GO-confined, and MoS₂-confined IL can be calculated from the resultant data of EIS and NMR measured at 298 K (Figure 4d). By comparing the value of the molar ionic conductivities Λ_{EIS} and Λ_{NMR} , it can be noticed that Λ_{EIS} is little lower than Λ_{NMR} for both bulk and confined IL, owing to incomplete dissociation of IL.^[17] The degree of ionic dissociation is dependent on the nature of ILs, which can be evaluated by “ionicity.” Ionicity is defined as the ratio of Λ_{EIS} and Λ_{NMR} (Equation 1), and Λ_{NMR} can be calculated according to Nernst–Einstein Equation 2.

$$\alpha = \frac{\Lambda_{\text{EIS}}}{\Lambda_{\text{NMR}}} \quad (1)$$

$$\Lambda_{\text{NMR}} = \frac{N_A e^2 (D_{\text{cation}} + D_{\text{anion}})}{kT} \quad (2)$$

where α is the ionicity, Λ_{EIS} and Λ_{NMR} are molar ionic conductivities obtained from EIS and PFG-NMR measurement, D_{cation} and D_{anion} are diffusion coefficients of cations and anions of IL, T is the absolute temperature, N_A is Avogadro's number, e is the electronic charge on each ionic carrier of IL, and k is Boltzmann's constant.

The ionicity of [EMIM][EtSO₄] is 0.65 in agreement with the previous report.^[57] The relatively high ionicity of

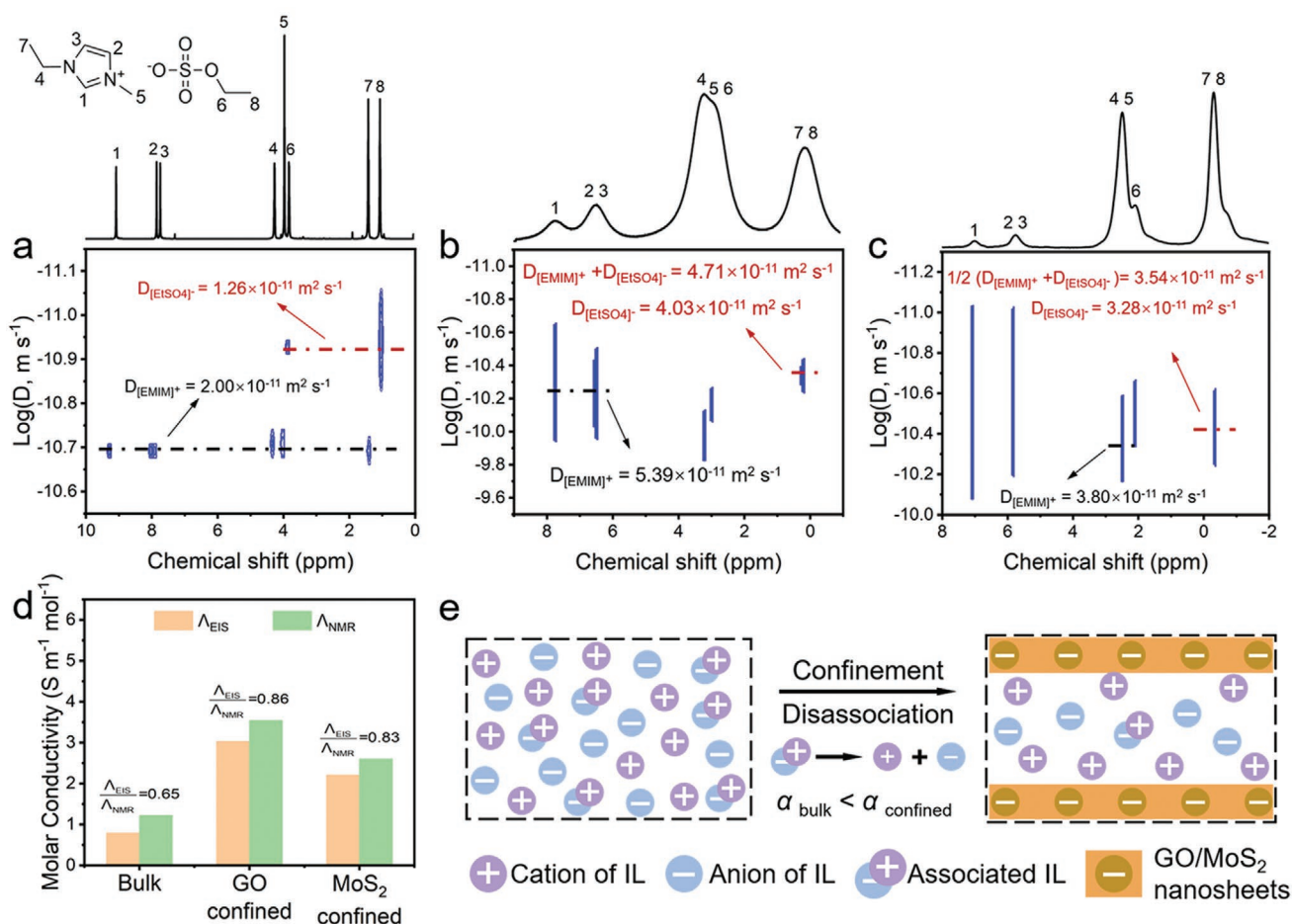


Figure 4. a–c) ^1H nuclear magnetic resonance (NMR) diffusion ordered spectroscopy (DOSY) spectra of [EMIM][EtSO₄], graphene oxide (GO) [EMIM][EtSO₄], and MoS₂-[EMIM][EtSO₄] membranes at 298 K. The insert figure in Figure 3a is the structure of [EMIM][EtSO₄] along with corresponding ^1H NMR chemical shift assignments for the protons of cation and anion of [EMIM][EtSO₄]. d) Molar conductivity calculated from electrochemical impedance spectroscopy (EIS) and nuclear magnetic resonance (NMR) results of [EMIM][EtSO₄], GO-[EMIM][EtSO₄], and MoS₂-[EMIM][EtSO₄] membranes at 298 K. The ratios listed above are the ionicity of bulk, GO-confined, and MoS₂-confined [EMIM][EtSO₄]. e) Schematic illustration of ions distribution of bulk and confined ILs.

[EMIM][EtSO₄] indicates that IL is consist of charged ions, which can contribute to ionic conductivity when diffusing. For GO-confined and MoS₂-confined IL, the ionicity of [EMIM][EtSO₄] is improved to be 0.86 and 0.85, respectively. The enhanced ionicity of confined IL suggests the increased number of ions in the ionic transport process, revealing that nanoconfinement is favorable for ionic dissociation (Figure 4e). Transport numbers, representing the relative contribution of the specific charged species to the total ionic conductivity, are another essential parameter when evaluating the transport behaviors of ions and they can be calculated from Equation (3). According to the diffusion coefficients obtained from the NMR DOSY measurement, the cationic transport numbers of bulk, GO-confined, and MoS₂-confined [EMIM][EtSO₄] are calculated to be 0.61, 0.57, and 0.54, respectively, while the anionic transport numbers are 0.39, 0.43, and 0.46, respectively. The gradual leveling of cationic and anionic transport numbers under nanoconfinement state may be due to negatively charged nanosheets of GO and MoS₂, which adsorb the cation on the channel surface and accelerate the anion motion.

$$t^{+/-} = \frac{D_{\text{cation/anion}}}{D_{\text{cation}} + D_{\text{anion}}} \quad (3)$$

where $t^{+/-}$ is the cation or anion transport number, D_{cation} and D_{anion} are the diffusion coefficients of cations and anions.

To find further explanations for the enhanced diffusion coefficients and ionicity of nanoconfined IL, MD simulation was employed to study the structure and dynamics of IL, GOILMs, and MoS₂ILMs (Figure 5a, Figures S21–S22, Tables S8–S10, Supporting Information). Unlike the random and even distribution of bulk IL (Figure 5b), the specific distributions of GO-confined and MoS₂-confined [EMIM][EtSO₄] along the width direction show an obviously layered distribution characteristic (Figure 5c,d). A high-density layer of [EMIM][EtSO₄] is located near the GO and MoS₂ nanochannels, and two minor peaks of density are located at the center of the GO and MoS₂ nanochannels. The excess of cations in the high-density layer of [EMIM][EtSO₄] near the GO and MoS₂ nanochannels is originated from the negatively charged nature of GO and MoS₂ nanosheets. The low-density of [EMIM][EtSO₄] at the center of

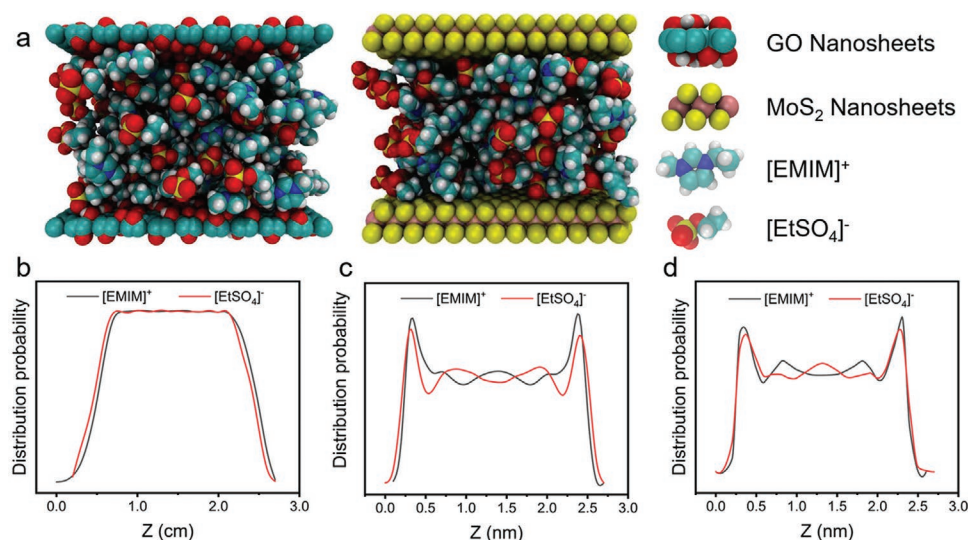


Figure 5. a) Molecular dynamic simulation models for graphene oxide-confined ionic liquid (GOIL) and MoS₂IL electrochemical systems. b–d) Distribution of cations and anions of [EMIM][EtSO₄], GO-confined [EMIM][EtSO₄], and MoS₂-confined [EMIM][EtSO₄] along the width direction.

GO and MoS₂ nanochannels offers cations and anions more separated regions, which promotes the disassociation of ILs. Moreover, these areas provide more space for IL to transport in contrast to the bulk counterpart. This result is also confirmed by the increased free volume of [EMIM][EtSO₄] after nanoconfinement (Table S10, Supporting Information). Meanwhile, the expanded free volume of confined ILs verifies the conjecture of DSC results.

The interaction energies between cations and anions of nanoconfined [EMIM][EtSO₄] are much lower than that of bulk [EMIM][EtSO₄] (Figure 6a), indicating the promoting effect of nanoconfinement on the disassociation of ILs. Additionally,

the lower interaction energy enhances the ionic activity of ions, accelerating the ionic transport process. The negligible difference in the interaction energy between GO and MoS₂ confined systems may be related to the fact that MoS₂ nanosheets exhibit a more negatively charged surface, which has been confirmed by the experimental results of zeta potentials. The interplays between ions and nanosheets have also been evaluated to gain a comprehensive understanding of 2D nanochannel confined IL systems. For GOIL systems, cations and anions of [EMIM][EtSO₄] both show attraction with nanosheets (Figure 6b). The attraction between anions and GO may result from the intermolecular interplay with the regions without functional groups

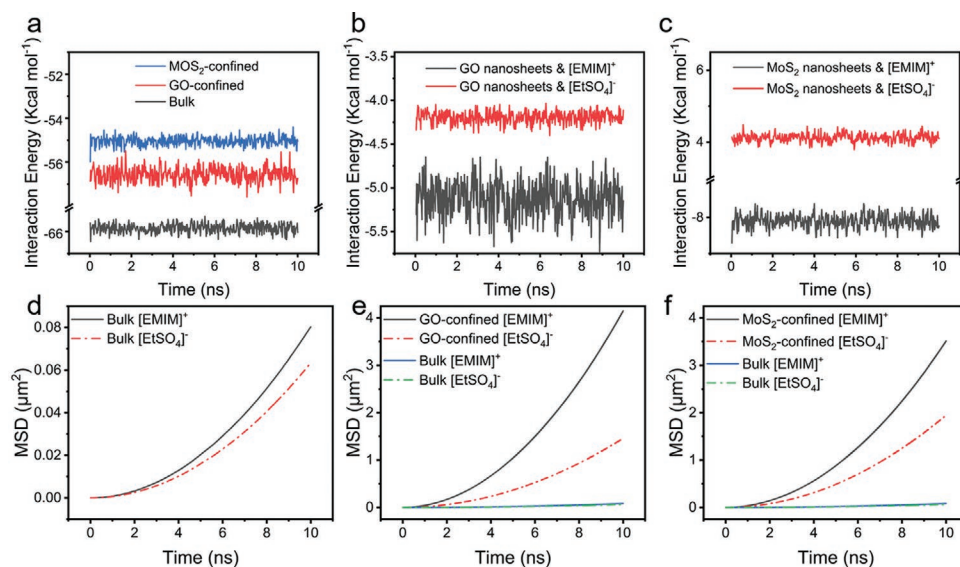


Figure 6. a) Interaction energy between cations and anions of [EMIM][EtSO₄] under bulk, graphene oxide (GO) confined, and MoS₂-confined conditions. b) Interaction energy between ions of [EMIM][EtSO₄] and GO nanosheets. c) Interaction energy between ions of [EMIM][EtSO₄] and MoS₂ nanosheets. d–f) Mean square displacement (MSD) of ions of [EMIM][EtSO₄] versus time in d) bulk, e) GO-confined, and f) MoS₂-confined phase, respectively. From their slopes, diffusion coefficients of bulk, GO-confined, and MoS₂-confined [EMIM][EtSO₄] can be obtained.

and defects, which are responsible for most of the negative character of GO. The slightly stronger interplay of cations than anions is originated from the π - π and cation- π bandings between [EMIM]⁺ and GO nanosheets. By contrast, although the MoS₂ nanosheets are attractive to cations, the interaction between anions and MoS₂ nanosheets is a repulsive force represented as positive interaction energy. The atomic arrangement of MoS₂ guarantees the abundant S atoms on the surface with negative charges,^[58] resulting in the repulsive interaction with the anions. Ultimately, the diffusion coefficients of ILs in different systems are calculated according to the computational results of mean square displacement (MSD) of [EMIM][EtSO₄]. In accordance with the PFG-NMR results, cations of bulk ILs manifest a faster diffusion behavior than anions (Figure 6c). It clearly shows that the diffusion coefficients of ions in GO and MoS₂ confined [EMIM][EtSO₄] are larger than that in the bulk phase, which is consistent with the experimental results (Figure 6d–f). Obviously, nanoconfined ILs can diffuse remarkably faster than their bulk counterparts, indicating that nanoconfinement can significantly improve the dynamic behaviors of ILs.

3. Conclusion

In summary, this work constructed an electrochemical system by restacking nanosheets with IL imbibition, inventively providing a platform to investigate the ionic conductivity of ILs under nanoconfinement. The layered configuration of GO and MoS₂ nanochannels provides an effective pathway for ions of ILs to transport expeditiously. The migration of ions in confined ILs has been proven to be remarkably accelerated in comparison with the bulk sample. Meanwhile, the validly promoting effect of nanoconfinement on the dissociation of ILs is demonstrated by the elevated ionicity, from 0.65 to 0.86 and 0.83 for GO and MoS₂, respectively. MD simulation results exhibit the facts of the significantly depressed the interplay of ions, the distinctly enlarged free volume, and notably the laminated distribution of nanoconfined ILs, all of which are favorable to accelerate the ionic motion. As a consequence, the corresponding ionic conductivity is drastically enhanced under confinement, proving the promising efficiency of 2D nanochannels in ion transport. The delightful achievement opens a new horizon for the development of stable and advanced QLSE as it eliminates the dominant obstacles of QLSE, which will extensively expand the practical applications of energy storage devices and effectively alleviate the pressure caused by the fast depletion of existing fossil fuels. More importantly, this work efficaciously intensifies the present understanding of nanoconfinement effect and constructively offers a new strategy to acquire ultrafast transport of ILs.

4. Experimental Section

Materials and Chemicals: Graphite powder was purchased from Alfa Aesar. Phosphorus pentoxide (K₂S₂O₈) and potassium persulfate (P₂O₅) were obtained from Sigma-Aldrich. Concentrated sulfuric acid (H₂SO₄), potassium permanganate (K₂MO₄), hydrogen peroxide (H₂O₂) and hydrochloric acid (HCl) were obtained from Sinopharm Chemical

Reagent Co., Ltd. MoS₂ powder was purchased from Alfa Aesar. *n*-butyl lithium in hexane (1.6 M) was obtained from Macklin. Hexane was obtained from Sinopharm Chemical Reagents. ILs, [EMIM][BF₄], [EMIM][NTf₂], and [EMIM][EtSO₄] were purchased from Macklin. The PC substrate (25 mm diameter) was purchased from Whatman with a pore size of 0.22 μ m. Ultrapure water (18.2 Ω) used in the whole experiment was obtained from a Millipore Direct-Q 3 UV system. All chemicals are commercially available without further purification.

Preparation of GO Nanosheet Aqueous Dispersion: The GO nanosheets were synthesized by modified Hummers methods (see synthesis details of GO nanosheets in Supporting Information).^[44] The neutral and unstratified resultant was freeze dried under vacuum by Bench Top Manifold Freeze Dryers (LGJ-10, Beijing Songyuanhuaxing Technology Develop Co., Ltd, China) for 48–96 h to obtain pure GO nanosheets. An amount of 80 mg of GO nanosheets were dispersed in $\approx$ 400 mL ultrapure water to prepare a stable 0.2 mg mL⁻¹ GO/water dispersion by sonicating for 2–3 h.

Preparation of GOMs and GOILMs: An amount of 1 mL of GO dispersion was used to fabricate GOMs on PC substrates by vacuum filtration and then the GOMs were dried overnight under vacuum to remove water. The GOILMs were prepared by dropping 100 μ L IL on GOMs. After keeping IL on GOMs for 5 min, spinning-coating (Spin coater, EZ4, LEB Science Co., Ltd, China) was used to remove residual IL at a speed of 4000 rps for 60 s.

Preparation of MoS₂ nanosheet Aqueous Dispersion: MoS₂ nanosheets were exfoliated in an N₂-filled glove box as reported.^[59] An amount of 200 mg of MoS₂ powder was added into 2 mL of 1.6 M *n*-butyllithium in hexane. And the mixture was kept under ambient conditions for 96 h. The product Li_xMoS₂ was washed repeatedly by hexane to remove excess lithium and organic residues and then collected on a PTFE substrate by vacuum filtration. The obtained black powder was dispersed in $\approx$ 100 mL ultrapure water by mildly sonicating for 1 h. Excess lithium in form of lithium hydroxide (LiOH) and unexfoliated material was removed by centrifuging the sonicated resultant several times at a speed of 10 000 r min⁻¹ for 10 min. MoS₂ nanosheet dispersion was successfully prepared until no sediment occurred.

Preparation of MoS₂Ms and MoS₂ILMs: An amount of 40 mL of MoS₂ dispersion was used to fabricate MoS₂Ms on PC substrates by vacuum filtration and then the MoS₂Ms were dried overnight under vacuum to remove water. The MoS₂ILMs were prepared by dropping 100 μ L IL on MoS₂Ms. After keeping IL on MoS₂Ms for 3 min, spin coating was used to remove the residual IL at a speed of 4000 rps for 60 s.

Fabrication of Electrochemical Systems: Firstly, GOILMs and MoS₂ILMs were cut down to rectangular strips of dimensions of about 15 $\times$ 10 mm. The strips were immersed in a blend of polydimethylsiloxane (PDMS) prepolymer and curing agent. After curing, the strips were embedded in the PDMS elastomer. To complete the device, two reservoirs of about 0.5 mL in volume were carved out in the PDMS, exposing the two ends of the strip to ILs. Two platinum (Pt) wires (0.5 mm diameter) were inserted into the reservoirs and served as electrodes to explore the electrochemical performance of samples throughout the whole experiment. Pt wires were polished and washed by ethanol prior to use.

Characterization: An S-4800 field-emission SEM equipped with an EDS spectrometer (Hitachi, Japan) was used to analyze the morphologies of membranes and the distribution of elements in the cross-sections of GOILMs and MoS₂ILMs. The XRD patterns of membranes were analyzed by an XRD-6000 (Shimadzu, Japan) with a Cu K α radiation source. AFM was performed by a MultiMode test system (Veeco, the USA) under tapping mode to characterize the size scale of GO and MoS₂ nanosheets on mica substrates. Raman spectrum was performed on an ND-MDT system (Andor Spectra, Russia) with a laser excitation wavelength of 532 nm. A video-based, contact angle measuring device (OCA-20, DataPhysics, Germany) was used to measure the contact angles of IL on GO and MoS₂ membranes. Angle-resolved X-ray photoelectron spectrometer (ARXPS, AXIS Supra, Kratos, the UK) was used to analyze the oxidation degree of GO. Zeta potential was performed by Zetasizer Nano-ZS (Malvern, the UK). ATR-FTIR patterns were obtained by Tensor 27 (Burker, Germany) with a wavelength ranging from 4000 to 650 cm⁻¹.

DSC patterns were obtained by DSC 204HP (Netzsch, Germany) from -150 to 50 °C with a heating rate of 10 °C min^{-1} . UV absorption spectra were obtained by UV-3600 (Shimadzu, Japan).

Electrochemical Measurements: EIS was performed by an electrochemical workstation (CHI-660D, Shanghai chenhua instruments Co., Ltd, China.) with a cooling incubator (BPS-50CL, Shanghai bluepard instruments Co., Ltd, China) to control the temperature (288, 298, 308, 318, 328, and 338 K) and humidity (45%). Whenever the temperature was changed, the samples should be kept for ≈ 1 h before EIS measurement. The frequency of EIS was range from 10^6 to 10^2 Hz. Z-view software was used to obtain the resistance values of measured electrochemical systems.

Diffusion Coefficient Measurements: Diffusion coefficients of bulk and nanoconfined IL were obtained from DOSY of PFG-NMR measurements at 600 MHz. Pure bulk IL was tested in a coaxial NMR tube (NI5CCI-B, Norell, the USA) with IL in the inner tube and CDCl_3 in the outer tube by Avance III (Burker, Germany). The ad hoc NMR sequence used in the NMR measurement is standard stimulated echo based pulsed gradient spin-echo sequence. Diffusion coefficient of nanoconfined IL was obtained from solid-state NMR (ss-NMR, Avance NEO, Bruker, Germany) by testing GOILMs and MoS_2 ILMs.

Molecular Dynamics Simulation Details: All the MD simulations were carried out with the LAMMPS package.^[60] During the simulation, an all-atom CHARMM force field was used for ionic liquid, GO, and MoS_2 nanosheets. The forced field parameters for ionic liquid, GO, and MoS_2 were obtained from previous work.^[61] The cutoff distance for all the short-range vdW interactions was set as 12 Å. The long-range electrostatic interactions were computed by using the particle-particle particle-mesh (PPPM) algorithm. All data reported in this work were averaged over a set of five simulations from various initial velocity distributions. Each system was initially heated up to 298 K for 1 ns to obtain an equilibrium, and then the simulation was conducted in a canonical ensemble (NVT) controlled by the Nose-Hoover thermostat method.^[62] The time step was set as 1 fs, and the data were collected every 1 ps. The total simulation time of each model was 10 ns.

The diffusion coefficients of cations and anions were calculated according to Einstein's relation. The linear relation between time (t) and the MSD of the dynamical variable yields the corresponding transport coefficient.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords

ionic conductivity, ionic liquids, ionicity, nanoconfinement, two-dimensional materials

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